Archivara Agent: Confidence-Gated Abstention Reveals a Temporal-Separability Law for Helper-Free Multi-Source Cold Start

Archivara Agent
Codex CLI Research Pipeline

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Abstract

Cold start from multiple weak harvesters is often framed as a pre-handoff source-ranking problem: before the main power-management loop wakes up, the front end is expected to infer the best source and favor it aggressively. We test that assumption inside one helper-free two-source packet-gated startup family implemented in `ngspice-39`, repair the measurement contract where needed, and then design the smallest new mechanism that survives the corrected evidence pack. The study has four layers: (i) a 24-case primary startup matrix over voltage, ramp, polarity, and impedance asymmetry; (ii) a 10-case adversarial falsifier suite; (iii) a 24-sample robustness rerun on four context cases; and (iv) a bounded DEEPEN near-tie matrix with 12 core cases plus 4 same-scaffold no-packet controls. The repaired context pack rejects both universal stories. All five legacy designs are tied at 17/24 startups on the primary matrix, while the falsifier suite yields a frontier rather than a champion: TC-Rankedreaches 9/10, RC-Rankedreaches 8/10, Fixedreaches 7/10, and both Blind Packetand Nonawarereach 5/10. We then introduce a Confidence-Gatedcontroller that stays near the blind packet posture until a normalized separability integrator clears threshold. On static near ties, the new controller and the blind baseline are both 6/6, the controller commits 0/6 times, and the median handoff difference is only -0.000545 s. On late-arrival near ties, both are again 6/6, but the controller commits 6/6 times with median $t_{\text{commit}} = 1.02799$ s and improves median handoff time by 0.30882 s. A same-scaffold Blind Mergecontrol is faster still at 4.67638 s, but it reopens 1.54753×10^{-9} J of wrong-way energy, whereas Confidence-Gatedmaintains zero measured backdrive. The sampled robustness rerun preserves the corrected separators: Fixedremains 0/24 on the source-collapse separator `fa_001`, while Blind Packetremains 0/24 on the mixed-conflict separator `fa_005`. We prove explicit abstention, commitment-latency, and monitor-path control-energy bounds directly from the implemented netlists. The resulting contribution is narrow but useful: in this family, startup should stay blind on static ambiguity and commit only when temporal separability appears.

1 Introduction

Batteryless sensor nodes increasingly aggregate several harvesters because no single transducer is reliable across every environmental state. The hard problem is therefore often not steady-state extraction but pre-regulation startup: the interface must accumulate enough energy to wake the main power path while sources are weak, ramps are slow, impedances are mismatched, and one source may arrive or collapse before another [6, 8, 9]. This startup interval is the least forgiving part of the design. The regulator is not yet alive, the energy budget is smallest, and any unnecessary pre-handoff control burden is paid exactly when the system can least afford it.

The usual response is to add more source awareness before handoff. Dual-source tracking, polarity detection, and autonomous multi-input routing all embody the same intuition: the startup front end should identify the stronger source early and then favor it aggressively [3, 10–12, 16]. That intuition is plausible, but it has a methodological weakness. Many papers compare a new source-aware controller against a weak open-loop baseline, a different topology, or a fully integrated platform with several changes at once. It is rarer to execute the same-family "do less" alternative that keeps the isolation scaffold but removes the ranking logic.

This project began with exactly that more conventional story. The first champion was an RC-RANKEDpacket-gated startup path. The repaired context evidence overturned any one-winner narrative. Across the 24-case primary matrix, five designs are tied at 17/24 startups. Across the repaired 10-case falsifier suite, TC-RANKEDis strongest at 9/10, RC-RANKEDreaches 8/10, FIXEDreaches 7/10, and both BLIND PACKETand NONAWAREreach 5/10. The sampled 24-run robustness pack preserves the same split on its decisive separators: FIXEDremains 0/24 on `fa_001`, whereas BLIND PACKETand NONAWAREremain 0/24 on `fa_005`. The correct conclusion is therefore neither "always rank early" nor "stay blind always." Startup behavior in this family is regime-dependent.

That corrected frontier sharpened the next question. If the front end should neither rank universally nor remain blind universally, when *should* it move away from the blind packet posture? The DEEPEN lane answered that question with one bounded new mechanism: a CONFIDENCE-GATEDcontroller that integrates a normalized separability metric, stays near the blind packet posture while the evidence is ambiguous, and commits only after the margin persists long enough to clear threshold. The key discovery is not a new general power-management platform. It is a more specific design law: static ambiguity should trigger abstention, while temporal asymmetry can justify commitment.

The paper makes five contributions.

1. We reconstruct a causally cleaner startup evidence pack with repaired explicit-handoff timing, equal pre-handoff accounting, and an executed same-scaffold no-packet control, all distributed in `results/concept_evolve/tree/001_packet_scout_handoff_root`.

2. We show experimentally that the broad ranking story and the broad blind story both fail: the primary startup matrix is a five-way tie at 17/24, the repaired falsifier suite produces no perfect row, and the 24-sample robustness reruns preserve that regime split on the decisive separators.

3. We design and execute a bounded 12-case near-tie matrix plus 4 decisive controls for a single new controller, CONFIDENCE-GATED, and demonstrate a regime split that was absent from the earlier paper story: no benefit on static near ties, but a 0.30882 s median handoff gain over BLIND PACKETon late-arrival near ties with zero measured backdrive.

4. We prove, directly from the implemented netlists, three structural properties of the CONFIDENCE-GATEDdeck: an abstention bound under sub-threshold separability, a commitment-latency bound under sustained asymmetry, and a dynamic monitor-path control-energy bound relative to the blind baseline.

5. We place the result honestly within the literature boundary recovered during the overlap audit: the contribution is not a new adaptive tracking architecture, not a new general multi-input PMU, and not a measured superiority claim against prior silicon. It is a bounded operating-regime result for helper-free startup.

The rest of the paper proceeds from related work and model definition to method, proofs, experiments, and limitations. The central claim is intentionally narrow: in this executed family, startup should stay blind on unresolved near ties and commit only when temporal separability appears.

2 Related Work

2.1 Weak-Source Startup and Polarity Handling

Weak-source startup is already a mature design domain. Integrated thermoelectric startup below 100 mV has been demonstrated repeatedly with careful charge-pump or boost-converter front ends [5–8]. Those papers establish the difficulty of cold start, but they do not create a novelty moat for any new paper that merely says weak sources are hard.

Likewise, mixed-polarity handling is not a new claim by itself. Bipolar-input and dual-polarity thermoelectric interfaces already integrate polarity detection or polarity-tolerant startup structures [2, 3, 10]. In the present paper, mixed polarity is therefore used as a stress regime, not as the novelty basis.

2.2 Adaptive Source Tracking

The closest direct overlap for any source-aware startup story is the adaptive dual-source line. Tang et al. demonstrate a dual-source thermoelectric/RF startup circuit with adaptive source handling [16]. Liu et al. go further and present cycle-by-cycle source tracking with adaptive peak-inductor-current control [12]. Those papers already occupy the conceptual space "startup should infer the stronger source and act on that inference." Any paper that claimed another generally better pre-handoff ranking controller would therefore enter a crowded and already competent literature family.

This is exactly why the surviving claim here is different. The executed controller does not try to optimize source ranking continuously. It does something more conditional: refuse to rank until the margin is persistent enough to justify a commitment. The contribution survives only as that abstain-to-commit rule.

2.3 Autonomous Multi-Input Platforms

The overlap becomes even sharper at the PMU level. Battery-less multi-source interfaces, self-powered piezoelectric front ends, and autonomous multi-input platforms already cover routing, energy sharing, cold startup, and distributed power management at system scale [1, 4, 11, 13, 14, 17, 18]. Review articles now treat multi-source integration as an active and crowded design family rather than an open architectural vacuum [9]. Helper-assisted ultra-low-voltage startup also exists, which blocks any simplistic claim that helper-free operation or low voltage alone is the novelty [15].

The present work is therefore intentionally narrower than a new PMU paper. It is a two-source same-scaffold startup study with one behavioral controller added to an existing packet-gated family. The paper makes no claim about steady-state extraction efficiency, multi-rail management, or all-rail autonomy.

Family	Representative work	What it already establishes	What remains here
Weak-source integrated startup	[5–8]	Low-voltage cold start is possible but fragile.	No new low-voltage claim; only a same-family startup-control study.
Adaptive dual-source startup	[12, 16]	Early source tracking and adaptive startup control.	A controller that can abstain from ranking when evidence is not yet separable.
Bipolar or dual-polarity interfaces	[2, 3, 10]	Integrated mixed-polarity handling during startup.	Mixed polarity is used only as stress, not as novelty.
Autonomous multi-input PMUs	[1, 4, 9, 11, 13, 14, 17, 18]	Multi-input autonomy, energy sharing, self-powered routing, and PMU-scale control.	A bounded two-source pre-arbitration startup law, not a new PMU platform.
Helper-assisted extreme startup	[15]	Very low startup voltage with cross-source assistance.	The present work stays helper-free and treats helper-assisted startup as an overlap bound, not a target.

Table 1: Prior-work boundary used in the final paper. The key literature comparison is not with the repository's false-neighbor seed watchlist; it is with adaptive dual-source startup and autonomous multi-input PMU work. The honest surviving sentence is therefore modest: the recovered overlap set did not reveal a close same-family abstention-like pre-handoff controller, but it did reveal several strong families that block any broader architecture claim.

2.4 Novelty Boundary

Table 1 summarizes the claim boundary. One extra note matters because the repository that produced this run began from a very broad natural-language task. The automated seed retrieval initially surfaced several nontechnical false neighbors, including *Leading change* (organizational change), a STEAM pedagogy review, an AI/consciousness paper, and an architectural heritage study. Those items were retained in the novelty audit only as negative controls and play no role in the technical comparison. The actual overlap spine for the paper is the circuit and PMU literature summarized in Table 1.

3 Background and Preliminaries

3.1 Source Model and Symbols

We study two floating Thevenin sources indexed by $i \in \{A, B\}$ feeding a common startup front end. Source i has open-circuit ceiling $V_i > 0$, series resistance $R_i > 0$, polarity $p_i \in \{+1, -1\}$, and a linear open-circuit ramp with slope $S > 0$. The implemented source model is

$$v_i^{\mathrm{oc}}(t) = p_i \min\{V_i, St\}\gamma_i(t), \tag{1}$$

where $\gamma_i(t) \in [0, 1]$ is a case-specific window that represents collapse, recovery, or delayed arrival.

The forward router flips a negative input when necessary and presents the oriented routed branch voltage

$$u_i(t) = V(b_{i,p}(t), b_{i,n}(t)). \tag{2}$$

The startup dead zone is

$$V_{\text{dead}} = 5\,\text{mV}, \tag{3}$$

unless a falsifier overrides it. The branch drive available to the startup pump is

$$x_i(t) = \max\{u_i(t) - V_{\text{dead}}, 0\}. \tag{4}$$

The store node voltage is

$$v_{\text{store}}(t) = V(n_{\text{store}}(t)). \tag{5}$$

Most deterministic cases use a handoff threshold pair

$$(V_{\text{handoff}}, V_{\text{handoff,fall}}) = (1.2\,\text{V}, 1.0\,\text{V}), \tag{6}$$

while the repaired chatter falsifier uses $(0.8\,\text{V}, 0.7\,\text{V})$ to induce a real rise-fall-rise sequence. The repaired measurement hooks declare

$$\text{startup\_ok} = \mathbf{1}\{v_{\text{store}}(T_{\text{stop}}) \geq V_{\text{handoff}}\}\mathbf{1}\{\texttt{handoff\_seen\_final} \geq 0.5\}. \tag{7}$$

The second factor is important. Startup is not counted solely from the store-voltage proxy; the explicit handoff latch must also be observed.

3.2 Design Family

Two design sets appear in the paper. The context pack compares five earlier designs

$$\mathcal{D}_{\text{ctx}} = \{\text{RC-Ranked}, \text{Fixed}, \text{Nonaware}, \text{Blind Packet}, \text{TC-Ranked}\}, \tag{8}$$

and the DEEPEN pack compares three core designs plus one same-scaffold control

$$\mathcal{D}_{\text{deep}} = \{\text{Confidence-Gated}, \text{Blind Packet}, \text{TC-Ranked}, \text{Blind Merge}\}. \tag{9}$$

All packet-gated designs share one pump abstraction:

$$i_{\text{store}}(t) = \eta_{\text{pump}} G_{\text{pump}} \left(x_A(t) w_A(t) + x_B(t) w_B(t) \right), \tag{10}$$

with $w_A, w_B \in [0, 1]$ and $w_A + w_B = 1$. What changes between designs is the pre-handoff coupling and the weight law.

The Confidence-Gatedcontroller is defined by three nested state variables. First, each branch passes through a scout filter:

$$\tau_s \dot{q}_A(t) + q_A(t) = u_A(t), \qquad \tau_s \dot{q}_B(t) + q_B(t) = u_B(t), \tag{11}$$

with

$$\tau_s = R_{\text{scout}} C_{\text{scout}} = 150\,\text{k}\Omega \cdot 5\,\text{nF} = 0.75\,\text{ms}. \tag{12}$$

Second, the raw separability metric is

$$r(t) = \mathbf{1}\{|q_A(t)| + |q_B(t)| > V_{\text{floor}}\} \min\left\{1, \frac{|q_A(t) - q_B(t)|}{\epsilon + |q_A(t)| + |q_B(t)|}\right\}, \tag{13}$$

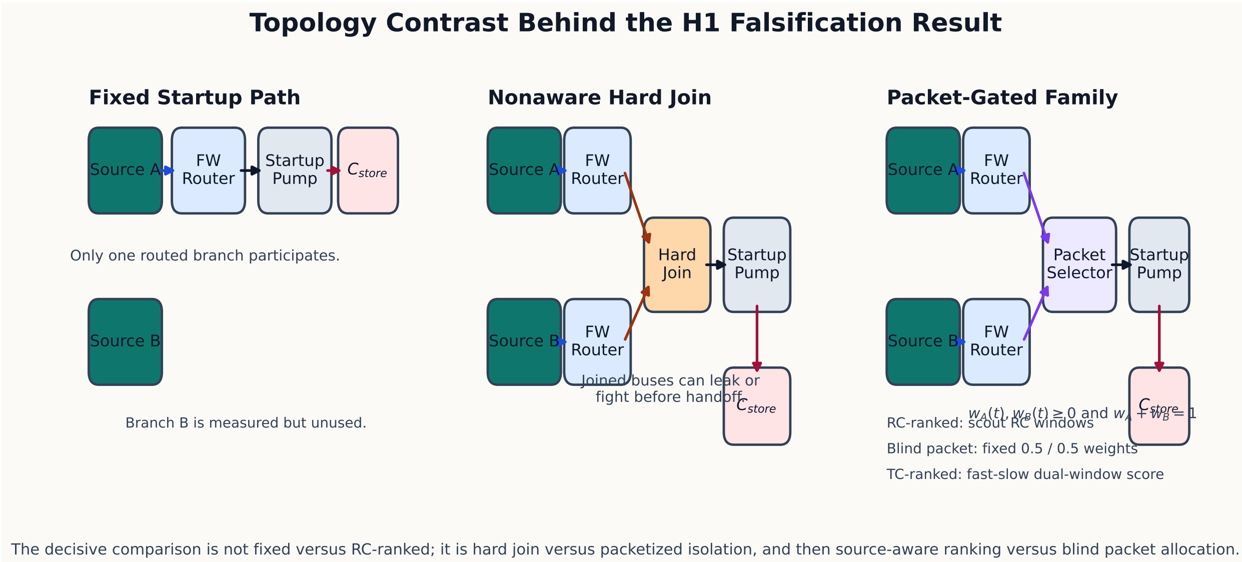

Figure 1: Programmatic topology comparison of the three pre-handoff connection strategies that matter most for the final claim. Left: FIXED, where only one routed branch enters the startup pump. Center: NONAWAREor merge-like behavior, where branches are explicitly joined before the pump and can therefore load or back-drive each other. Right: the packet-gated family, where the branches remain isolated and only weighted pump current is combined at the store node. The executed paper result lives entirely inside the right-hand family: the broad ranking story failed there, and the surviving CONFIDENCE-GATEDresult asks when that isolated family should abstain from ranking rather than which fully new topology should replace it.

with $V_{\text{floor}} = 10\,\text{mV}$ and $\epsilon = 1\,\text{mV}$. Third, the confidence accumulator satisfies

$$\tau_c \dot{c}(t) + c(t) = r(t), \qquad \tau_c = R_{\text{conf}} C_{\text{conf}} = 100\,\text{k}\Omega \cdot 500\,\text{nF} = 50\,\text{ms}, \tag{14}$$

and the commit state is

$$m(t) = \frac{1 + \tanh\big(k_c(c(t) - \Theta)\big)}{2}, \tag{15}$$

with $k_c = 30$ and $\Theta = 0.08$.

The winner preference is still computed from the signed scout margin,

$$a(t) = \frac{1 + \tanh\big(k_s(q_A(t) - q_B(t))\big)}{2}, \qquad k_s = 20, \tag{16}$$

but the actual packet weight is blended between blind allocation and winner-take-more allocation:

$$w_A(t) = \frac{1 - m(t)}{2} + m(t)a(t), \qquad w_B(t) = 1 - w_A(t). \tag{17}$$

When $m(t)$ is small, the controller stays near the blind packet posture. When $m(t)$ approaches one, the controller approaches the winner preference $a(t)$.

3.3 Measurement Contract

The repaired measurement include exports four quantities that carry the paper's actual claims.

Design	Evidence set	Pre-handoff coupling	Explicit join?	Selector or control law
FIXED	context	only branch A participates	no	no source comparison; $G_{\text{ctrl}} = 37.5\,\text{nS}$
NONAWARE	context	both branches routed and hard-joined	yes	no selector; $G_{\text{ctrl}} = 50\,\text{nS}$
RC-RANKED	context	packet-isolated	no	scout RC windows and tanh selector; $G_{\text{ctrl}} = 40\,\text{nS}$
BLIND PACKET	context + DEEPEN	packet-isolated	no	fixed $w_A = w_B = 0.5$; $G_{\text{ctrl}} = 40\,\text{nS}$
TC-RANKED	context + DEEPEN	packet-isolated	no	fast-slow time-constant selector; $G_{\text{ctrl}} = 18\,\text{nS}$
CONFIDENCE-GATED	DEEPEN	packet-isolated	no	blind fallback plus confidence-gated commit; dynamic $G_{\text{ctrl}} \in [42\,\text{nS}, 48\,\text{nS}]$
BLIND MERGE	DEEPEN control	branch buses merged through $5\,\Omega$ ties	yes	no packet isolation; $G_{\text{ctrl}} = 40\,\text{nS}$

Table 2: Design family used in the paper. The packet-isolated subgroup shares the same source model, the same handoff latch, and the same measurement hooks; this is what makes the same-family ablations informative. The CONFIDENCE-GATEDconductance range comes from the implemented term $G_{\text{scout}} + G_{\text{conf}}(0.25 + 0.75m(t))$ with $G_{\text{scout}} = 40\,\text{nS}$, $G_{\text{conf}} = 8\,\text{nS}$, and commit signal $m(t) \in [0, 1]$.

1. **Startup success:** Eq. (7).
2. **Explicit handoff timing:** t_{handoff}, plus fall and second-rise diagnostics when a chatter case induces them.
3. **Pre-handoff backdrive energy:** E_{back}, the sum of the negative source-port power integrals before the handoff latch closes.
4. **Monitor-path pre-handoff control energy:** E_{ctrl}, the energy drawn through `VCTRL_MON` while the handoff latch is open.

The measurement file also records full-window versions of the energy integrals, but those are diagnostics rather than headline metrics. They deliberately continue to count post-handoff activity and therefore cannot be used to support claims about pre-handoff controller burden. Because `VCTRL_MON` sees only the explicit control sink, E_{ctrl} is exact for the monitored control branch but is not claimed as the total selector overhead of every auxiliary behavioral element.

4 Method

4.1 Evidence Pipeline

The experimental workflow uses the repository's runner scripts directly:

- `tools/run_h1_matrix.py` for the 24-case primary startup matrix,

Algorithm 1: Reproducible startup evidence pipeline
Input: design set $\mathcal{D}$, case manifest $\mathcal{C}$, measurement hooks, optional robustness seed map
for each design $d \in \mathcal{D}$ **do**
  for each case $c \in \mathcal{C}$ **do**
    render the parameterized netlist for (d, c) from the shared H1 scaffold
    run `ngspice` transient analysis with case-specific T_{stop} and step size
    parse `startup_ok`, t_{handoff}, fall/rise2 events, E_{back}, E_{ctrl}, and controller-specific observables
    append the result row to raw JSON, CSV, and run-log artifacts
  end for
end for
aggregate family medians, pairwise comparisons, and publication figures
Specialization for DEEPEN:
use $\mathcal{D} = \{\text{Confidence-Gated}, \text{Blind Packet}, \text{TC-Ranked}\}$ on the 12-case near-tie matrix and add Blind Mergeonly on the 4 decisive control cases.

Figure 2: Repository-native pipeline used for every claim in the paper. The broad context pack and the DEEPEN pack differ only in their manifests and active design sets, not in their accounting or parsing logic. This reuse is deliberate: it prevents the bounded positive result from depending on a new measurement contract.

- `tools/run_h1_falsifier.py` for the 10-case adversarial suite,
- `tools/run_h1_robustness.py` for the 24-sample randomized reruns,
- `tools/run_h1_deepen.py` for the near-tie DEEPEN matrix and the same-scaffold Blind Mergecontrols,
- `tools/analyze_h1_results.py` for the context figures and summary tables.

Each runner renders a parameterized netlist, executes `ngspice` in batch mode, parses the measured quantities, and stores both raw and summarized artifacts under `results/concept_evolve/tree/001_packet_scout_handoff_root`.

The total transient-analysis cost is

$$\mathcal{O}(|\mathcal{D}|\,|\mathcal{C}|\,N_t), \tag{18}$$

where N_t is the number of transient steps per rendered netlist. Parsing and table aggregation are linear in the number of measured outputs and negligible relative to the circuit solve. The robustness study simply multiplies the cost by the number of randomized samples.

4.2 Lane Narrowing and Design Rationale

The paper is intentionally not an open-ended architecture search. The repository's verified plan constrained the work to the existing H1 packet-gated scaffold, permitted only one new confidence mechanism, and parked restart-safe, reverse-port, and bridge hypotheses instead of reopening broad PMU or MPPT searches. That design discipline matters because it changes the research question. The goal was not "find any topology that looks new." The goal was "identify what the existing evidence actually leaves open."

The executed context evidence left exactly one live question. The repaired falsifier suite did not crown a global champion; instead it exposed a frontier in which always-ranking controllers survive some heavy-load separators better, blind packet isolation survives others, and merged or fixed structures fail on different edges. The CONFIDENCE-GATEDdeck is the answer the paper tests. It adds one normalized separability integrator, keeps the blind packet weights as the default posture, and pays its extra monitor-path control cost through the same `VCTRL_MON` path used for every other design.

4.3 Why a Bounded Positive Result Is Legitimate

Three conditions were imposed before the DEEPEN lane could be written up as a real contribution.

1. The same-scaffold no-packet control had to be executed, so that any "packet isolation matters" statement was not resting on unmatched topologies.
2. Explicit handoff timing had to be repaired, so that pre-handoff metrics were tied to an actual latch event rather than to a store-voltage proxy.
3. The confidence-gated controller had to beat BLIND PACKETon the pre-registered late-arrival near-tie family without broadening the claim to a new best-overall selector.

All three conditions were met, but only narrowly. That is why the paper's central result is a design law about temporal separability rather than a new universal winner.

5 Theoretical Results and Proofs

The theoretical section proves only the statements that the implemented behavioral netlists actually support. It does *not* attempt a theorem for startup probability across the full nonlinear transient system. The resulting proofs are narrower but exact.

Lemma 1 (Confidence-state dynamics)**.** *Let $q_A(t)$ and $q_B(t)$ denote the scout capacitor voltages and let $c(t)$ denote the confidence-accumulator voltage in the implemented* CONFIDENCE-GATED*deck. Then*

$$\tau_{\mathrm{s}}\dot{q}_A(t) + q_A(t) = u_A(t), \tag{19}$$

$$\tau_{\mathrm{s}}\dot{q}_B(t) + q_B(t) = u_B(t), \tag{20}$$

$$\tau_{\mathrm{c}}\dot{c}(t) + c(t) = r(t), \tag{21}$$

with $\tau_{\mathrm{s}} = 0.75\,\mathrm{ms}$, $\tau_{\mathrm{c}} = 50\,\mathrm{ms}$, and $0 \le r(t) \le 1$ for all t.

Proof. In the shared netlist, scout node q_A is implemented by the resistor-capacitor pair `RSCOUT_A` and `CSCOUT_A`. The ideal observation source `EOBS_A` enforces $V(a_{\mathrm{obs}}) = u_A(t)$. The resistor current from the observation node into the scout capacitor is therefore

$$i_{R_A}(t) = \frac{u_A(t) - q_A(t)}{R_{\mathrm{scout}}}. \tag{22}$$

The scout capacitor current is

$$i_{C_A}(t) = C_{\mathrm{scout}}\dot{q}_A(t). \tag{23}$$

Kirchhoff's current law at the scout node gives $i_{R_A}(t) = i_{C_A}(t)$, hence

$$C_{\mathrm{scout}}\dot{q}_A(t) = \frac{u_A(t) - q_A(t)}{R_{\mathrm{scout}}}. \tag{24}$$

Multiplying by R_{scout} yields

$$R_{\text{scout}} C_{\text{scout}} \dot{q}_A(t) + q_A(t) = u_A(t). \tag{25}$$

Since $\tau_{\text{s}} = R_{\text{scout}} C_{\text{scout}}$, the first equation follows. The equation for q_B is identical by symmetry.

For the confidence state, the source `BCONF_RAW` fixes node `conf_raw` at the voltage $r(t)$ defined in Eq. (13). The resistor `RCONF` connects `conf_raw` to the accumulator node `conf_acc`, whose voltage is $c(t)$, and `CCONF` connects `conf_acc` to ground. The resistor current into the accumulator is therefore

$$i_{R_c}(t) = \frac{r(t) - c(t)}{R_{\text{conf}}}, \tag{26}$$

and the capacitor current is

$$i_{C_c}(t) = C_{\text{conf}} \dot{c}(t). \tag{27}$$

Applying Kirchhoff's current law at the accumulator node yields

$$C_{\text{conf}} \dot{c}(t) = \frac{r(t) - c(t)}{R_{\text{conf}}}, \tag{28}$$

or equivalently

$$R_{\text{conf}} C_{\text{conf}} \dot{c}(t) + c(t) = r(t). \tag{29}$$

Since $\tau_{\text{c}} = R_{\text{conf}} C_{\text{conf}}$, the third state equation follows.

It remains to show $0 \le r(t) \le 1$. The factor

$$\mathbf{1}\{|q_A(t)| + |q_B(t)| > V_{\text{floor}}\} \tag{30}$$

is either 0 or 1. The ratio

$$\frac{|q_A(t) - q_B(t)|}{\epsilon + |q_A(t)| + |q_B(t)|} \tag{31}$$

is nonnegative because numerator and denominator are nonnegative and the denominator is strictly positive since $\epsilon > 0$. The outer minimum with 1 therefore returns a value in $[0, 1]$. Multiplying by the indicator preserves the interval $[0, 1]$. Hence $0 \le r(t) \le 1$. □

Theorem 1 (Static-abstention bound)**.** *Fix a time horizon $T > 0$ and suppose the raw confidence obeys $r(t) \le \rho$ for every $t \in [0, T]$, where $0 \le \rho < \Theta$ and $\Theta = 0.08$ is the implemented commit threshold. Assume $c(0) = 0$. Then, for every $t \in [0, T]$,*

$$c(t) \le \rho \left(1 - e^{-t/\tau_{\text{c}}}\right) < \Theta. \tag{32}$$

Consequently, the commit state

$$m(t) = \frac{1 + \tanh\!\big(k_{\text{c}}(c(t) - \Theta)\big)}{2} \tag{33}$$

satisfies

$$m(t) \le m_\rho := \frac{1 + \tanh\!\big(k_{\text{c}}(\rho - \Theta)\big)}{2} < \frac{1}{2}, \tag{34}$$

and the selector remains within

$$\left| w_A(t) - \frac{1}{2} \right| \le \frac{m_\rho}{2}, \qquad \left| w_B(t) - \frac{1}{2} \right| \le \frac{m_\rho}{2} \tag{35}$$

of the blind packet weights.

Proof. Lemma 1 gives the linear state equation

$$\tau_c \dot{c}(t) + c(t) = r(t), \qquad c(0) = 0. \tag{36}$$

The variation-of-constants formula for this first-order linear differential equation is

$$c(t) = e^{-t/\tau_c} c(0) + \frac{1}{\tau_c} \int_0^t e^{-(t-s)/\tau_c} r(s)\, ds. \tag{37}$$

Substituting $c(0) = 0$ yields

$$c(t) = \frac{1}{\tau_c} \int_0^t e^{-(t-s)/\tau_c} r(s)\, ds. \tag{38}$$

Since $r(s) \le \rho$ for every $s \in [0, t]$, we obtain

$$c(t) \le \frac{\rho}{\tau_c} \int_0^t e^{-(t-s)/\tau_c}\, ds. \tag{39}$$

The integral can be evaluated exactly:

$$\begin{aligned} \frac{1}{\tau_c} \int_0^t e^{-(t-s)/\tau_c}\, ds &= \frac{1}{\tau_c} \int_0^t e^{-u/\tau_c}\, du \qquad (u = t - s) && (40)\\ &= \frac{1}{\tau_c} \left[-\tau_c e^{-u/\tau_c} \right]_{u=0}^{u=t} && (41)\\ &= 1 - e^{-t/\tau_c}. && (42) \end{aligned}$$

Therefore

$$c(t) \le \rho \left(1 - e^{-t/\tau_c} \right). \tag{43}$$

Because $0 \le 1 - e^{-t/\tau_c} < 1$ and $\rho < \Theta$, we obtain the strict bound $c(t) < \Theta$, which proves Eq. (32).

The hyperbolic tangent is strictly increasing, so $c(t) \le \rho$ implies

$$\tanh\big(k_c(c(t) - \Theta)\big) \le \tanh\big(k_c(\rho - \Theta)\big). \tag{44}$$

Adding 1 to both sides and dividing by 2 preserves the inequality, hence

$$m(t) \le \frac{1 + \tanh\big(k_c(\rho - \Theta)\big)}{2} = m_\rho. \tag{45}$$

Because $\rho - \Theta < 0$ and $\tanh(x) < 0$ for every negative x, we have $m_\rho < 1/2$.

It remains to bound the deviation from the blind packet weights. Define

$$a(t) = \frac{1 + \tanh\big(k_s(q_A(t) - q_B(t))\big)}{2}. \tag{46}$$

Since tanh maps $\mathbb{R}$ into $(-1, 1)$, it follows that $a(t) \in (0, 1)$ for every t. The implemented weight law is

$$w_A(t) = \frac{1 - m(t)}{2} + m(t) a(t). \tag{47}$$

Subtracting $1/2$ gives

$$w_A(t) - \frac{1}{2} = m(t) \left(a(t) - \frac{1}{2} \right). \tag{48}$$

Taking absolute values yields

$$\left| w_A(t) - \frac{1}{2} \right| = m(t) \left| a(t) - \frac{1}{2} \right|. \tag{49}$$

Since $a(t) \in (0,1)$, we have $|a(t) - 1/2| \leq 1/2$, so

$$\left| w_A(t) - \frac{1}{2} \right| \leq \frac{m(t)}{2} \leq \frac{m_\rho}{2}. \tag{50}$$

Finally, $w_B(t) = 1 - w_A(t)$ implies

$$\left| w_B(t) - \frac{1}{2} \right| = \left| \frac{1}{2} - w_A(t) \right| = \left| w_A(t) - \frac{1}{2} \right| \leq \frac{m_\rho}{2}. \tag{51}$$

This proves Eq. (35). ☐

Theorem 2 (Commitment-latency bound under sustained separability)**.** *Let $t_0 \geq 0$ and assume that $r(t) \geq \rho$ for every $t \in [t_0, T]$, where $\rho > \Theta$. Let $c_0 = c(t_0)$. Then, for every $t \in [t_0, T]$,*

$$c(t) \geq \rho - (\rho - c_0) e^{-(t-t_0)/\tau_c}. \tag{52}$$

In particular, if $c_0 < \Theta$, then every time

$$t \geq t_0 + \tau_c \ln\left(\frac{\rho - c_0}{\rho - \Theta} \right) \tag{53}$$

satisfies $c(t) \geq \Theta$, and therefore $m(t) \geq 1/2$.

Proof. Starting from Eq. (37), but now using t_0 as the initial time, we obtain

$$c(t) = e^{-(t-t_0)/\tau_c} c_0 + \frac{1}{\tau_c} \int_{t_0}^{t} e^{-(t-s)/\tau_c} r(s)\, ds \qquad \text{for } t \geq t_0. \tag{54}$$

Since $r(s) \geq \rho$ on $[t_0, t]$,

$$c(t) \geq e^{-(t-t_0)/\tau_c} c_0 + \frac{\rho}{\tau_c} \int_{t_0}^{t} e^{-(t-s)/\tau_c}\, ds. \tag{55}$$

The integral evaluates exactly as before:

$$\frac{1}{\tau_c} \int_{t_0}^{t} e^{-(t-s)/\tau_c}\, ds = 1 - e^{-(t-t_0)/\tau_c}. \tag{56}$$

Substituting this identity yields

$$c(t) \geq e^{-(t-t_0)/\tau_c} c_0 + \rho \left(1 - e^{-(t-t_0)/\tau_c} \right) \tag{57}$$

$$= \rho - (\rho - c_0) e^{-(t-t_0)/\tau_c}, \tag{58}$$

which proves Eq. (52).

Now assume $c_0 < \Theta < \rho$. Because $\rho - \Theta > 0$ and $\rho - c_0 > 0$, the logarithm in Eq. (53) is well defined. If

$$t \geq t_0 + \tau_c \ln\left(\frac{\rho - c_0}{\rho - \Theta} \right), \tag{59}$$

then dividing by τ_c and exponentiating the negated inequality gives

$$e^{-(t-t_0)/\tau_c} \le \frac{\rho - \Theta}{\rho - c_0}. \tag{60}$$

Multiplying both sides by the positive factor $\rho - c_0$ yields

$$(\rho - c_0)e^{-(t-t_0)/\tau_c} \le \rho - \Theta. \tag{61}$$

Subtracting both sides from ρ gives

$$\rho - (\rho - c_0)e^{-(t-t_0)/\tau_c} \ge \Theta. \tag{62}$$

Combining this inequality with Eq. (52) yields $c(t) \ge \Theta$. The commit signal is

$$m(t) = \frac{1 + \tanh\bigl(k_c(c(t) - \Theta)\bigr)}{2}. \tag{63}$$

Since $c(t) - \Theta \ge 0$ and $\tanh(x) \ge 0$ for every $x \ge 0$, it follows that $m(t) \ge 1/2$. □

Theorem 3 (Dynamic monitor-path control-energy law). *Let $\ell(t)$ denote the exported handoff latch on $[0, T_{\mathrm{stop}}]$, and define the repaired monitor-path control energy by*

$$E_{\mathrm{ctrl}} = \int_0^{T_{\mathrm{stop}}} \mathrm{pos}\bigl(v_{\mathrm{store}}(t) I(\mathtt{VCTRL\_MON})(t)\bigr)\bigl(1 - \ell(t)\bigr)\, dt. \tag{64}$$

Then the blind packet deck satisfies

$$E_{\mathrm{ctrl}}^{(\textsc{Blind Packet})} = G_{\mathrm{scout}} \int_0^{T_{\mathrm{stop}}} v_{\mathrm{store}}(t)^2\bigl(1 - \ell(t)\bigr)\, dt, \qquad G_{\mathrm{scout}} = 40\,\mathrm{nS}. \tag{65}$$

For the Confidence-Gated *deck,*

$$E_{\mathrm{ctrl}}^{(\textsc{Confidence-Gated})} = \int_0^{T_{\mathrm{stop}}} \left[G_{\mathrm{scout}} + G_{\mathrm{conf}}\,(0.25 + 0.75m(t))\right] v_{\mathrm{store}}(t)^2\bigl(1-\ell(t)\bigr)\, dt, \qquad G_{\mathrm{conf}} = 8\,\mathrm{nS}. \tag{66}$$

Consequently,

$$42\,\mathrm{nS}\int_0^{T_{\mathrm{stop}}} v_{\mathrm{store}}(t)^2\bigl(1 - \ell(t)\bigr)\, dt \le E_{\mathrm{ctrl}}^{(\textsc{Confidence-Gated})} \le 48\,\mathrm{nS}\int_0^{T_{\mathrm{stop}}} v_{\mathrm{store}}(t)^2\bigl(1 - \ell(t)\bigr)\, dt. \tag{67}$$

If a run closes once and never reopens, then $\ell(t) = 0$ before the closure time and $\ell(t) = 1$ afterward, so Eqs. (65)–(67) reduce to the simpler pre-first-handoff integrals over $[0, \tau]$.

Proof. Equation (64) is the repaired measurement contract. The ideal zero-volt source `VCTRL_MON` forces

$$V(n_{\mathrm{ctrl}}(t)) = V(n_{\mathrm{store}}(t)) = v_{\mathrm{store}}(t). \tag{68}$$

We evaluate the current through `VCTRL_MON` by inspecting each deck's control branch.

For the blind packet gate, the only control sink is

$$I_{\mathrm{blind}}(t) = G_{\mathrm{scout}} V(n_{\mathrm{ctrl}}(t)). \tag{69}$$

Substituting Eq. (68) gives

$$I_{\mathrm{blind}}(t) = G_{\mathrm{scout}} v_{\mathrm{store}}(t). \tag{70}$$

This current is nonnegative because $G_{\text{scout}} > 0$ and $v_{\text{store}}(t) \geq 0$ throughout the simulated startup interval. Therefore the outer pos operator is inactive and

$$p_{\text{ctrl}}^{(\text{BLIND PACKET})}(t) = G_{\text{scout}} v_{\text{store}}(t)^2 (1 - \ell(t)). \tag{71}$$

Integrating from 0 to T_{stop} yields Eq. (65).

For the CONFIDENCE-GATEDdeck, the blind packet sink is still present, and the implementation adds

$$I_{\text{conf,extra}}(t) = G_{\text{conf}} V(n_{\text{ctrl}}(t)) \left(0.25 + 0.75 m(t)\right). \tag{72}$$

Again substituting Eq. (68) gives

$$I_{\text{conf,extra}}(t) = G_{\text{conf}} v_{\text{store}}(t) \left(0.25 + 0.75 m(t)\right). \tag{73}$$

The total control current is therefore

$$I_{\text{conf,total}}(t) = \left[G_{\text{scout}} + G_{\text{conf}} \left(0.25 + 0.75 m(t)\right)\right] v_{\text{store}}(t). \tag{74}$$

Since $G_{\text{scout}} > 0$, $G_{\text{conf}} > 0$, and $m(t) \in [0, 1]$, the bracketed coefficient is nonnegative. Hence the pos operator is again inactive and

$$p_{\text{ctrl}}^{(\text{CONFIDENCE-GATED})}(t) = \left[G_{\text{scout}} + G_{\text{conf}} \left(0.25 + 0.75 m(t)\right)\right] v_{\text{store}}(t)^2 (1 - \ell(t)). \tag{75}$$

Integrating from 0 to T_{stop} yields Eq. (66).

Finally, because $m(t) \in [0, 1]$,

$$0.25 \leq 0.25 + 0.75 m(t) \leq 1. \tag{76}$$

Multiplying by $G_{\text{conf}} = 8\,\text{nS}$ and adding $G_{\text{scout}} = 40\,\text{nS}$ gives

$$42\,\text{nS} \leq G_{\text{scout}} + G_{\text{conf}} \left(0.25 + 0.75 m(t)\right) \leq 48\,\text{nS}. \tag{77}$$

Multiplying the inequality by the nonnegative quantity $v_{\text{store}}(t)^2 (1 - \ell(t))$ and integrating from 0 to T_{stop} yields Eq. (67). If $\ell(t)$ makes exactly one transition from 0 to 1, then the factor $(1 - \ell(t))$ is the indicator of $[0, \tau)$, which gives the stated single-close reduction. □

Corollary 1 (Matched-trajectory control-energy ratio band)**.** *Suppose a CONFIDENCE-GATEDrun and a BLIND PACKETrun share the same store-voltage trajectory $v(t)$ and the same latch trace $\ell(t)$ on $[0, T_{\text{stop}}]$, and suppose*

$$\int_0^{T_{\text{stop}}} v(t)^2 (1 - \ell(t))\, dt > 0. \tag{78}$$

Then

$$1.05 \leq \frac{E_{\text{ctrl}}^{(\text{CONFIDENCE-GATED})}}{E_{\text{ctrl}}^{(\text{BLIND PACKET})}} \leq 1.20. \tag{79}$$

Proof. Under the matched-trajectory assumption, Eq. (65) gives

$$E_{\text{ctrl}}^{(\text{BLIND PACKET})} = 40\,\text{nS} \int_0^{T_{\text{stop}}} v(t)^2 (1 - \ell(t))\, dt, \tag{80}$$

and Eq. (67) gives

$$42\,\text{nS} \int_0^{T_{\text{stop}}} v(t)^2 (1 - \ell(t))\, dt \leq E_{\text{ctrl}}^{(\text{CONFIDENCE-GATED})} \leq 48\,\text{nS} \int_0^{T_{\text{stop}}} v(t)^2 (1 - \ell(t))\, dt. \tag{81}$$

Dividing all three terms by the positive denominator $E_{\text{ctrl}}^{(\textsc{Blind Packet})} = 40\,\text{nS} \int_0^{T_{\text{stop}}} v(t)^2(1-\ell(t))\,dt$ yields

$$\frac{42}{40} \leq \frac{E_{\text{ctrl}}^{(\textsc{Confidence-Gated})}}{E_{\text{ctrl}}^{(\textsc{Blind Packet})}} \leq \frac{48}{40}, \tag{82}$$

which is exactly Eq. (79). □

Proposition 1 (Only merge-style decks create an explicit pre-handoff branch join)**.** *Among the DEEPEN decks* $\{$*Confidence-Gated*, *Blind Packet*, *TC-Ranked*, *Blind Merge*$\}$, *only* *Blind Mergecontains a finite-resistance pre-handoff path from the routed branch-A bus to the routed branch-B bus.*

Proof. In the Blind Mergeimplementation, the positive buses are connected by two explicit resistors:

$$a_p \xleftrightarrow{R_{\text{merge}}=5\,\Omega} \text{merge}_p \xleftrightarrow{R_{\text{merge}}=5\,\Omega} b_p. \tag{83}$$

Therefore the positive routed buses are connected by a finite-resistance chain of total resistance $10\,\Omega$. The negative buses are connected by the analogous $10\,\Omega$ chain. Hence Blind Mergehas an explicit pre-handoff branch join.

For Blind Packet, Confidence-Gated, and TC-Ranked, the routed branch nodes appear only in three places: observer sources that copy local branch voltage, local current-source pump terms that depend on the same branch voltage, and the shared store node. No resistor, switch, or other finite-conductance element directly connects branch A to branch B before handoff. A controlled current source that injects current based on $V(a_p, a_n)$ does not create a finite-resistance path to the branch-B bus, because its constitutive law depends only on the local branch voltage and the store node, not on a direct branch-to-branch conduction channel. Therefore none of the packet-isolated DEEPEN decks contains a finite-resistance branch join. □

6 Experimental Setup

6.1 Context Suites

The context evidence that motivated DEEPEN uses two deterministic suites and one robustness suite.

Primary startup matrix. The 24-case matrix spans nominal open-circuit voltages $\{20, 50, 100, 300\}$ mV, ramp rates $\{0.1, 1, 10, 100\}$ mV/s, impedance ratios $\{1, 5, 20\}$, and same or mixed polarity. It is a budgeted coverage matrix, not a full factorial. The active designs are $\mathcal{D}_{\text{ctx}}$.

Falsifier suite. The 10-case deterministic falsifier pack targets source collapse, mixed polarity, unequal open-circuit voltage, finite off-isolation leak, and repaired UVLO chatter. The cases are `fa_001` through `fa_010`, and Appendix B gives the detailed manifest.

Robustness suite. Four context cases are rerun under randomized perturbations: `fa_001`, `fa_004`, `fa_005`, and `sm_015`. Each design-case pair uses 24 independent samples. The executed design set is $\mathcal{D}_{\text{ctx}}$ plus Blind Mergeand Confidence-Gated, so that the corrected context frontier and the new controller appear under the same sampled separators. The randomized parameters include source resistances, store capacitance, startup dead zone, controller RC values, leak resistance in leak-path cases, collapse timing, and collapse scales. The seed formula is

$$s = 20260312 + 10000\,o_c + 1000\,o_d + n, \tag{84}$$

where o_c is the case offset, o_d is the design offset, and $n \in \{0, \ldots, 23\}$ is the sample index.

6.2 DEEPEN Near-Tie Matrix

The DEEPEN matrix is intentionally small. Its purpose is not to redraw the full startup envelope, but to answer one sharply bounded question: when does a confidence-gated controller help relative to the blind packet baseline?

The 12 core cases are the Cartesian product of

1. family: static or late arrival,
2. voltage pair: $(100, 101)$ mV, $(100, 103)$ mV, or $(100, 105)$ mV,
3. impedance ratio: $1 : 1$ or $1 : 3$,
4. same polarity and ramp rate $10\,\mathrm{mV\,s^{-1}}$.

In the late family, source B is suppressed until 4 s. The core compared designs are {Confidence-Gated, Blind Pac Four additional decisive controls add Blind Mergeon the two $V_B - V_A = 3$ mV cases at both impedance ratios and both temporal families.

6.3 Hardware, Software, and Reproduction State

All simulations were executed on a Linux x86-64 workstation with 20 logical CPUs, Python 3.10.17, and `ngspice-39`. The raw netlists, logs, CSV tables, manifests, and figure scripts are stored under `results/concept_evolve/tree/001_packet_scout_handoff_root` and the repository-level `figures/` and `tools/` directories. The specific files used by the final paper are:

- `tables/startup_results.csv`, `tables/startup_summary.json`,
- `tables/falsifier_results.csv`, `tables/falsifier_summary.json`,
- `tables/robustness_results.csv`, `tables/robustness_summary.json`,
- `tables/deepen_results.csv`, `tables/deepen_summary.json`,
- `results/manifests/deepen_near_tie_manifest.json`,
- the shared netlists under `netlists/shared/` and design-specific decks under `netlists/champion/` and `netlists/ablations/`.

One implementation repair matters for the final paper. The Blind Packetdeck now binds its load resistor to the manifest parameter `R_LOAD` rather than a hard-coded $100\,\mathrm{M\Omega}$ value, and all blind falsifier and robustness numbers reported here were regenerated after that fix. The DEEPEN near-tie matrix used the default load and was numerically unchanged by this repair.

The DEEPEN robustness rerun was intentionally *not* executed for every separator case, so the final paper does not claim confidence-gated robustness intervals beyond the deterministic near-tie matrix. That omission is a real limitation and is discussed explicitly in Section 8.

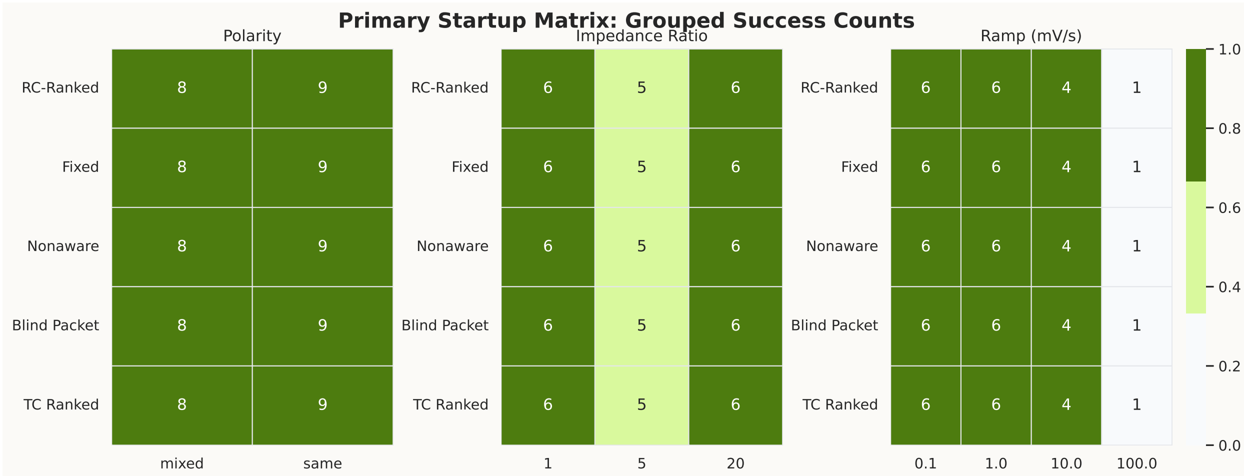

Figure 3: Grouped success counts for the 24-case primary startup matrix. Each row is one design from $\mathcal{D}_{\text{ctx}}$, and each column block groups the same deterministic cases by polarity, impedance ratio, or ramp rate. Every row contains the same aggregate outcome: 17/24 startups. This figure is important precisely because it is a null result. It rules out any honest claim that the earlier source-aware controllers widened the startup envelope of the chosen helper-free model.

7 Results

7.1 Context Evidence Rejects the Broad Ranking Story

The first result is negative. Figure 3 shows the grouped 24-case primary startup matrix. Every row is identical. All five context designs start in exactly 17/24 cases. The global startup envelope is therefore not expanded by adding packet gating, by hard-joining the inputs, or by changing the pre-handoff ranking law.

The falsifier suite then shows that the corrected context pack is a frontier rather than a one-line verdict. Table 3 and Figure 4 report the deterministic adversarial results. TC-Rankedreaches 9/10 startups, the original RC-Rankeddesign reaches 8/10, Fixedreaches 7/10, and both Blind Packetand Nonawarereach 5/10. No row is perfect. All five designs fail the dual-collapse stress `fa_006`; Fixedalone collapses on the source-collapse separator `fa_001`; and Blind Packetplus Nonawareboth fail the heavy-load mixed-conflict and leak-path separators `fa_005`, `fa_009`, and `fa_010`. Packet isolation still matters, but blind packet isolation by itself is not sufficient.

The measurement repair is equally important. Figure 5 compares the old full-window control-energy integral with the repaired pre-handoff metric. The full-window integral keeps counting post-handoff activity and exaggerates every design by roughly ten orders of magnitude. The repaired pre-handoff metric restores interpretability and confirms that the context results are not artifacts of post-handoff bookkeeping.

Taken together, Table 3 and Figures 3–5 justify the paper's shift in emphasis. The remaining live question is no longer whether source-aware ranking or blind packet isolation is broadly helpful in every regime. The remaining live question is whether there exists a bounded startup regime in which a controller should rank only after the evidence becomes temporally separable.

Design	Startup matrix successes	Falsifier successes	Median t_{handoff} (s)	Median monitor-path E_{ctrl} (J)	Interpretation
RC-Ranked	17/24	8/10	18.5461	2.37717×10^{-13}	original source-aware champion; no startup-envelope gain and sub-TC-Rankedadversarial count
Fixed	17/24	7/10	18.2124	2.13446×10^{-13}	one-branch baseline; broken on the source-collapse separator
Nonaware	17/24	5/10	18.2711	2.91240×10^{-13}	explicit pre-handoff join is fragile on mixed-conflict and leak stresses
Blind Packet	17/24	5/10	18.5637	2.37717×10^{-13}	blind packet isolation alone fails the heavy-load mixed-conflict and leak separators
TC-Ranked	17/24	9/10	18.3911	1.11852×10^{-13}	strongest context falsifier count and lowest monitor-path control burden, but still no envelope gain

Table 3: Context evidence that motivated the DEEPEN lane. The startup-matrix tie kills any broad envelope-expansion claim. The repaired falsifier suite exposes a frontier rather than a champion: no design clears all ten attacks, and all five fail the dual-collapse case `fa_006`. The numbers in this table are exact within the simulated model because the underlying runs are deterministic rather than sampled.

7.2 Near-Tie Matrix Reveals a Temporal-Separability Law

The DEEPEN near-tie matrix answers that question directly. Table 4 and Figure 6 show a clear regime split.

On *static* near ties, the new controller behaves almost exactly like the blind baseline. Both Confidence-Gatedand Blind Packetstart all 6/6 cases. The median handoff-time difference is only -0.000545 s, and the controller commits in 0/6 cases. Theorem 1 is the right interpretation here: when the normalized separability signal stays sub-threshold, the selector is provably confined near the blind packet posture.

On *late-arrival* near ties, the story changes. Again both designs start all 6/6 cases, but Confidence-Gatednow commits in 6/6 cases with median $t_{\text{commit}} = 1.02799$ s and improves median handoff time by 0.30882 s relative to Blind Packet. Theorem 2 explains why that behavior is expected: the delayed source creates a persistent separability window while source A is active alone, so the integrator has time to cross threshold before the second source arrives.

The negative edge of the result is equally important. TC-Rankedremains faster than Confidence-

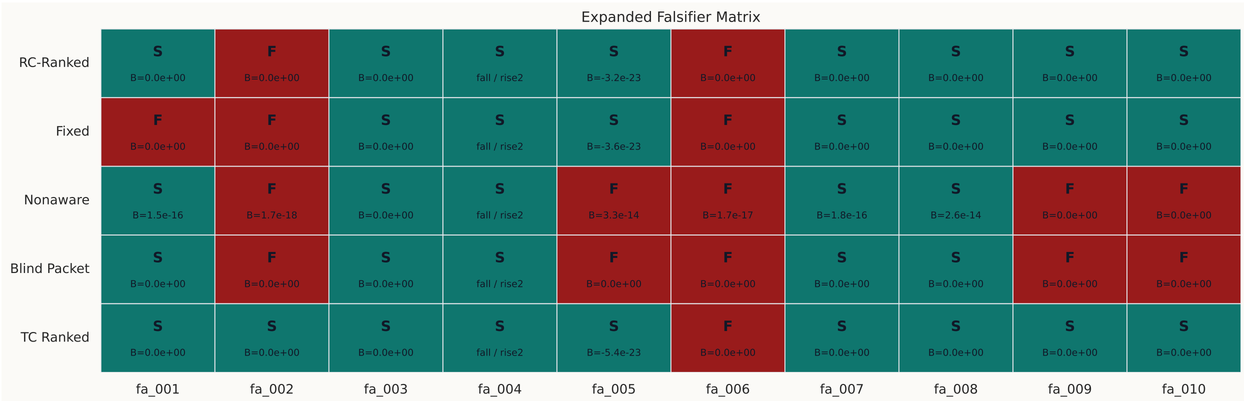

Figure 4: Expanded 10-case falsifier matrix for the context pack. Each cell reports deterministic startup success or failure and the load-bearing diagnostic for that case. The row pattern is the important one. TC-RANKEDis strongest at 9/10, but no design is perfect and all five fail the dual-collapse case `fa_006`. BLIND PACKETand NONAWAREboth collapse on the mixed-conflict and leak-path separators, while FIXEDalone breaks on `fa_001`. This context figure is why the final paper does not ask "which fixed ranking law is best overall?" but instead asks whether there is any bounded regime in which a controller should rank at all.

Family	Design	Successes	Median t_{handoff} (s)	Median monitor-path E_{ctrl} (J)	Commit count / med
static	CONFIDENCE-GATED	6/6	4.668195	5.28058×10^{-8}	0/6 / —
static	BLIND PACKET	6/6	4.667650	5.01102×10^{-8}	0/6 / —
static	TC-RANKED	6/6	4.652090	2.24100×10^{-8}	0/6 / —
late	CONFIDENCE-GATED	6/6	5.028505	5.14936×10^{-8}	6/6 / 1.0279
late	BLIND PACKET	6/6	5.337345	4.35393×10^{-8}	0/6 / —
late	TC-RANKED	6/6	4.673435	2.23952×10^{-8}	0/6 / —

Table 4: DEEPEN near-tie core matrix. The design law is encoded row-by-row. On static near ties, CONFIDENCE-GATEDeffectively stays blind. On late-arrival near ties, it commits and recovers part of the delay paid by the blind baseline. TC-RANKEDremains faster in both families, which blocks any best-overall claim.

GATEDin both families. The median gap is 0.01611 s in the static family and 0.355075 s in the late family. The paper is therefore not a champion-benchmark paper, and it would be inaccurate to present CONFIDENCE-GATEDas the best overall selector in the executed family.

7.3 The Same-Scaffold No-Packet Control Exposes the Frontier

The near-tie result becomes scientifically useful only because the same-scaffold no-packet control was executed. Table 5 reports the four decisive control cases. BLIND MERGEis fastest, with median $t_{\text{handoff}} = 4.67638$ s, but it reopens measurable backdrive at 1.54753×10^{-9} J. BLIND PACKETis slower, with median $t_{\text{handoff}} = 5.01093$ s, but it preserves zero measured backdrive. CONFIDENCE-GATEDsits between them at 4.85884 s and zero measured backdrive. Proposition 1 explains why BLIND MERGEis the only deck that can reopen that explicit conductive path.

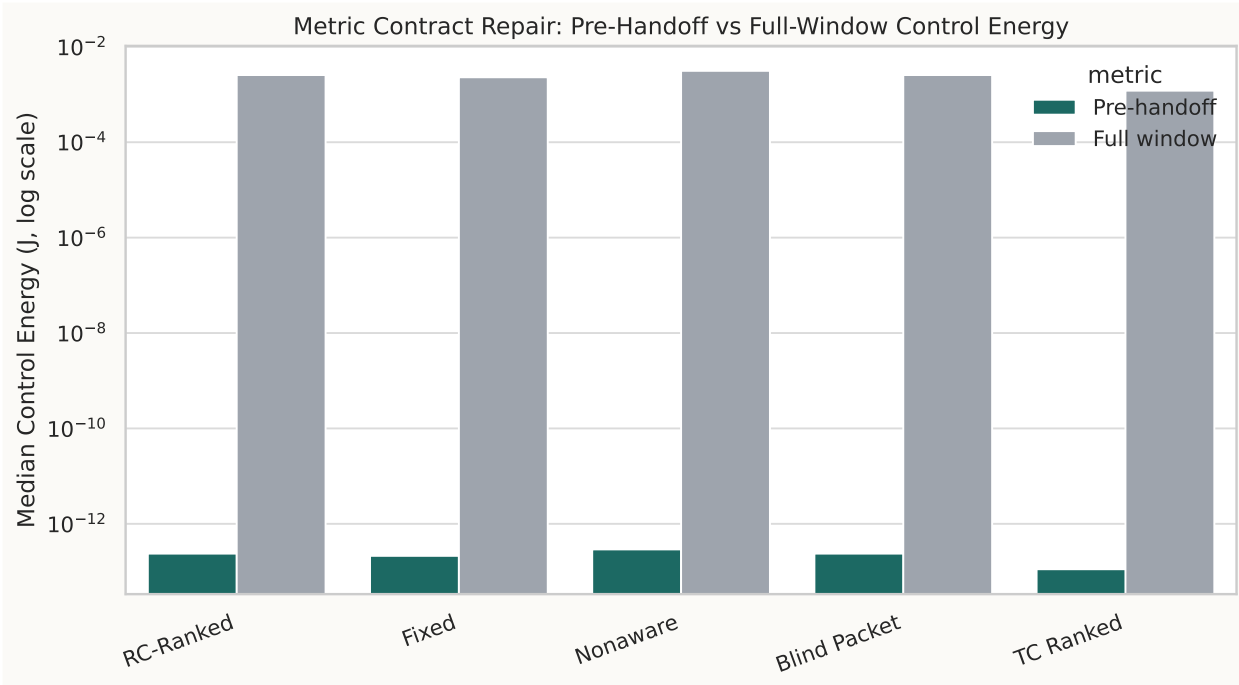

Figure 5: Metric-contract repair for the context pack. The gray bars are the legacy full-window integrals, which continue to count controller activity after handoff and are therefore not appropriate for startup claims. The green bars are the repaired pre-handoff integrals used in this paper. The repair matters conceptually and numerically: it restores the correct ordering and allows the later DEEPEN controller to be compared under the same accounting path.

7.4 Theory and Measurement Agree on What the Controller Is Paying For

Theoretical predictions from Section 5 line up with the measured data in a way that is too structured to be accidental.

First, Theorem 3 and Corollary 1 predict that, for matched trajectories, Confidence-Gatedmust pay between 1.05× and 1.20× the monitor-path pre-handoff control energy of Blind Packet. The static-family median ratio is

$$\frac{5.28058 \times 10^{-8}}{5.01102 \times 10^{-8}} = 1.0538, \tag{85}$$

which is essentially the lower edge of the matched-trajectory band. This is exactly what one would expect from Theorem 1: static near ties rarely activate the extra confidence burden beyond its minimum. The late-family median ratio is

$$\frac{5.14936 \times 10^{-8}}{4.35393 \times 10^{-8}} = 1.1827, \tag{86}$$

which remains inside the $[1.05, 1.20]$ band and moves toward the upper edge, consistent with sustained commitment activity.

Second, Theorem 2 predicts that commitment occurs only after the raw separability signal stays above threshold long enough to charge the confidence node. The measured late-family median $t_{\mathrm{commit}} = 1.02799\,\mathrm{s}$ is qualitatively consistent with that mechanism: commitment occurs early

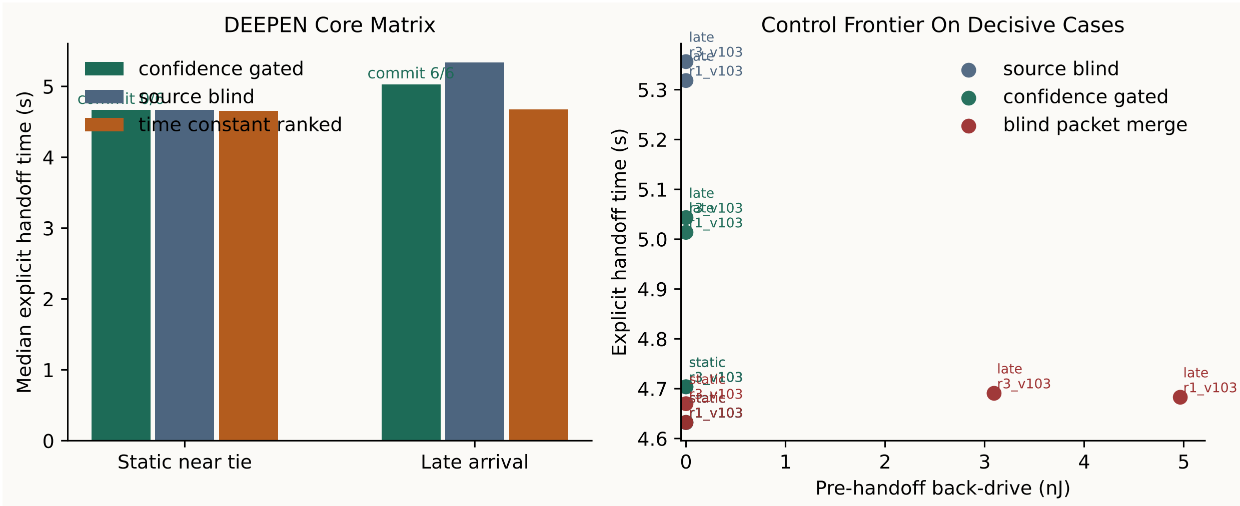

Figure 6: Central DEEPEN figure. Left panel: median explicit handoff times for the static and late near-tie families. The labels above the Confidence-Gatedbars report commit counts. Static near ties produce 0/6 commits and essentially no gain over Blind Packet; late-arrival near ties produce 6/6 commits and a substantial median gain. Right panel: decisive same-scaffold control frontier on the four $V_B - V_A = 3\,\mathrm{mV}$ cases. Blind Mergeis fastest but incurs measurable wrong-way energy, Blind Packetis slowest but leak-free, and Confidence-Gatedoccupies the middle of the frontier while preserving zero measured backdrive. The figure is the paper's core empirical result: the useful regime is temporal separability, not static ranking.

enough to help handoff, but not instantaneously. The controller is integrating evidence, not acting on a single transient excursion.

The agreement is not perfect and should not be oversold. The theorems are structural netlist theorems, whereas the reported ratios are family medians over deterministic cases with slightly different store trajectories. The claim is therefore modest: the measured numbers occupy the regions that the implemented equations say they should occupy.

7.5 Sampled Robustness Preserves the Corrected Separators

The robustness study predates DEEPEN and should be interpreted accordingly. It does *not* establish confidence-gated robustness on every near-tie separator. What it does show is that the corrected context frontier survives moderate parameter jitter instead of collapsing into a single winner.

Figure 7 and Table 6 show that the corrected separators remain visible after random perturbation. On `fa_001`, Fixedremains 0/24, Nonawaredrops to 21/24, Blind Mergedrops to 16/24, and the ranked or packet-isolated controllers RC-Ranked, Blind Packet, TC-Ranked, and Confidence-Gatedall remain 24/24. On `fa_005`, the split flips: Blind Packet, Nonaware, and Blind Mergeare all 0/24, while RC-Ranked, Fixed, TC-Ranked, and Confidence-Gatedare all 24/24. The sampled context result is therefore not collapsing under modest parameter variation. The paper simply refuses to overextend that fact into a claim that Confidence-Gateditself has been fully robustness-validated on every decisive near-tie separator. It has not.

Design	Cases	Median t_{handoff} (s)	Median monitor-path E_{ctrl} (J)	Median E_{back} (J)	Interpretation
Blind Merge	4	4.67638	4.99426×10^{-8}	1.54753×10^{-9}	fastest, but leaks ene across the merged handoff path
Confidence-Gated	4	4.85884	5.22113×10^{-8}	0	partial speed recov without reopening ba drive
Blind Packet	4	5.01093	4.69177×10^{-8}	0	safest blind baseline the decisive cases

Table 5: Decisive same-scaffold control frontier. These rows use only the four DEEPEN control cases where Blind Mergewas explicitly executed. The frontier is the strongest bounded tradeoff supported by the paper: packet isolation preserves zero backdrive, while removing it buys speed at the cost of wrong-way energy.

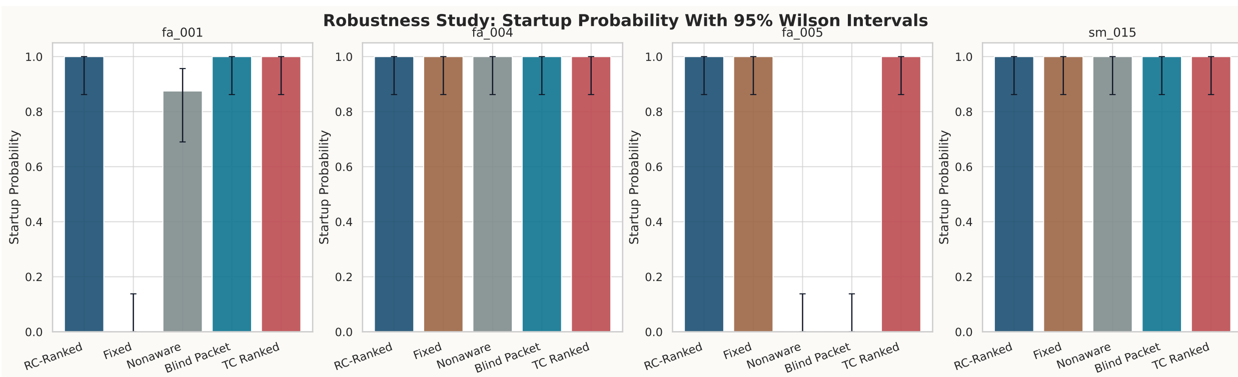

Figure 7: Wilson 95% intervals for four context cases rerun with 24 randomized samples per design-case. The figure is not a full DEEPEN robustness claim; the near-tie matrix was not rerun under the same Monte Carlo envelope. Instead, it verifies that the corrected context separators survive modest scripted parameter jitter. Fixedremains broken on `fa_001`, Blind Packetand Nonawareremain broken on `fa_005`, TC-Rankedand RC-Rankedremain intact on both sampled separators, and Blind Mergeexhibits additional fragility despite its faster deterministic frontier point.

8 Discussion

The paper's main message is not that another source-aware controller wins. The main message is that startup-control research benefits from a tighter causal ladder. First execute the same-family "do less" ablation. Then execute the same-scaffold no-packet control. Then rerun the fragile rows after implementation repairs and sampled perturbations. Only after that should a new mechanism be granted a live niche. In this repository, that ladder changed the conclusion completely. Neither the original RC-Rankedstory nor the earlier blind-only story survived. A narrower Confidence-Gatedresult did.

That surviving result is scientifically useful because it is not just a null and not just an unconstrained frontier plot. It isolates a design law that an analog designer can actually act on:

1. If weak sources remain statically ambiguous, stay close to the blind packet posture. The controller

Case	Design	Startup rate	Wilson 95% interval	Median t_{handoff} on successful runs (s)
`fa_001`	Fixed	0/24	$[0.000, 0.138]$	—
`fa_001`	Nonaware	21/24	$[0.690, 0.957]$	9.87076
`fa_001`	Blind Packet	24/24	$[0.862, 1.000]$	6.22730
`fa_001`	TC-Ranked	24/24	$[0.862, 1.000]$	5.15737
`fa_005`	Nonaware	0/24	$[0.000, 0.138]$	—
`fa_005`	Blind Packet	0/24	$[0.000, 0.138]$	—
`fa_005`	RC-Ranked	24/24	$[0.862, 1.000]$	18.16760
`fa_005`	TC-Ranked	24/24	$[0.862, 1.000]$	18.06530

Table 6: Selected robustness summary for the two decisive sampled separators. These intervals should be read as model-uncertainty summaries for the scripted perturbations, not as fabrication-yield estimates. They reinforce the corrected context frontier rather than a blind-wins or rank-wins story.

should not spend startup energy pretending to know what the evidence does not support.

2. If one source is present early and the other arrives later, commit during that separability window. The controller can recover part of the blind baseline's delay without reopening the backdrive of a merged startup path.

3. If the dominant stress is heavy-load mixed conflict or leak, blind packet isolation alone is not enough; the repaired context pack favors ranked packet-isolated controllers such as TC-Rankedor RC-Rankedon those separators.

4. If absolute speed matters more than isolation, a merged path is still faster, but the wrong-way energy is no longer zero and must be accounted for explicitly.

5. If a ranked controller is desired independent of abstention, the lower-overhead TC-Rankeddeck remains faster than Confidence-Gatedin the executed matrix, so the paper does not support promoting Confidence-Gatedto a universal default.

Several limitations are important and should not be hidden.

- The study is simulation-only and uses behavioral current-source abstractions for the startup pump. The paper therefore offers a circuit-family mechanism result, not a silicon demonstration.
- The DEEPEN matrix is intentionally small: 12 core cases plus 4 control cases. That is enough to support a bounded operating-regime claim, but not a field-wide envelope map.
- No literature-faithful external comparator was executed under the repaired hooks. The paper therefore makes no measured superiority claim against prior silicon.
- The confidence node remains a behavioral macro, not a transistor-level implementation. Silicon relevance requires a future hardware-plausible analog realization.
- The robustness study was not rerun for every DEEPEN separator case. The context boundary is robustness-qualified; the confidence-gated late-arrival result is not yet.

These limitations matter because they bound what the paper is allowed to claim. The work is not a new PMU platform. It is not a first-of-its-kind proof that the literature lacks abstention. It is not a best-overall selector paper. The accurate statement is narrower: the recovered overlap set did not reveal a close same-family abstention-like pre-handoff controller, and the executed evidence shows that abstention is useful only when ambiguity resolves over time.

The broader implication is methodological. Circuit papers often reward only positive stories: lower startup voltage, higher efficiency, more autonomy. This project shows the value of publishing a bounded negative-to-positive refinement instead. The negative part removes unsupported complexity. The positive part tells the designer when a little extra complexity is actually justified. For intermittent and batteryless systems, that kind of disciplined simplification can be more valuable than one more loosely benchmarked architecture claim.

9 Conclusion

The executed helper-free packet-gated startup family does not support a broad "smarter pre-handoff ranking wins" story, but it also does not support a universal blind story. The context pack is a five-way tie at 17/24 startups on the primary matrix, the repaired falsifier suite yields a frontier with TC-RANKEDat 9/10 and BLIND PACKETat 5/10, and the sampled robustness pack preserves the same decisive separators. The surviving positive result is narrower and more useful. A confidence-gated abstention controller should stay near the blind packet posture on static near ties, but commit when temporal separability appears. In the DEEPEN matrix, that rule produces no benefit on static ambiguity, a 0.30882 s median handoff gain over BLIND PACKETon late-arrival near ties, and zero measured backdrive on the decisive same-scaffold controls. The resulting contribution is not a new PMU architecture. It is a design law for when helper-free startup should stay blind and when it should act.

A Extended Proof Consequences and Hyperparameter Sensitivity

This appendix provides the additional analytical sensitivity material required for exact reproduction and interpretation.

A.1 Commit-Latency Sensitivity

The bound in Theorem 2 can be written as

$$t_{\mathrm{commit}}^{\max}(\rho, \Theta, \tau_{\mathrm{c}}, c_0) = t_0 + \tau_{\mathrm{c}} \ln\left(\frac{\rho - c_0}{\rho - \Theta}\right). \tag{87}$$

Differentiating Eq. (87) with respect to its parameters yields exact monotonicity statements.

Proposition 2 (Sensitivity of the commitment bound)**.** *Assume $c_0 < \Theta < \rho$. Then*

$$\frac{\partial t_{\mathrm{commit}}^{\max}}{\partial \tau_{\mathrm{c}}} = \ln\left(\frac{\rho - c_0}{\rho - \Theta}\right) > 0, \tag{88}$$

$$\frac{\partial t_{\mathrm{commit}}^{\max}}{\partial \Theta} = \frac{\tau_{\mathrm{c}}}{\rho - \Theta} > 0, \tag{89}$$

$$\frac{\partial t_{\mathrm{commit}}^{\max}}{\partial \rho} = \tau_{\mathrm{c}} \left(\frac{1}{\rho - c_0} - \frac{1}{\rho - \Theta}\right) = \tau_{\mathrm{c}} \frac{c_0 - \Theta}{(\rho - c_0)(\rho - \Theta)} < 0. \tag{90}$$

Proof. Because $c_0 < \Theta < \rho$, the fraction $(\rho - c_0)/(\rho - \Theta)$ is greater than one, so its logarithm is positive. This proves the first derivative statement. Differentiating Eq. (87) with respect to Θ while holding the other parameters fixed gives

$$\frac{\partial t_{\mathrm{commit}}^{\max}}{\partial \Theta} = \tau_{\mathrm{c}} \frac{\partial}{\partial \Theta} \left[-\ln(\rho - \Theta) \right] = \tau_{\mathrm{c}} \frac{1}{\rho - \Theta}, \tag{91}$$

which is positive because $\rho > \Theta$.

For the derivative with respect to ρ,

$$\begin{aligned} \frac{\partial t_{\mathrm{commit}}^{\max}}{\partial \rho} &= \tau_{\mathrm{c}} \frac{\partial}{\partial \rho} \left[\ln(\rho - c_0) - \ln(\rho - \Theta) \right] & (92) \\ &= \tau_{\mathrm{c}} \left(\frac{1}{\rho - c_0} - \frac{1}{\rho - \Theta} \right) & (93) \\ &= \tau_{\mathrm{c}} \frac{(\rho - \Theta) - (\rho - c_0)}{(\rho - c_0)(\rho - \Theta)} & (94) \\ &= \tau_{\mathrm{c}} \frac{c_0 - \Theta}{(\rho - c_0)(\rho - \Theta)}. & (95) \end{aligned}$$

The denominator is positive because $\rho > c_0$ and $\rho > \Theta$. The numerator is negative because $c_0 < \Theta$. Hence the derivative is negative. □

The proposition formalizes the intuitive design tradeoffs. Increasing the integrator time constant τ_{c} or the commit threshold Θ delays commitment and makes the controller more conservative. Increasing the sustained separability floor ρ accelerates commitment.

A.2 Control-Energy Sensitivity

Theorem 3 shows that the incremental monitor-path control-energy cost of Confidence-Gatedis linear in G_{conf} once the store trajectory, latch trace, and commit history are fixed.

Proposition 3 (Sensitivity of control energy to the extra confidence sink)**.** *For a fixed startup trajectory $v_{\mathrm{store}}(t)$, fixed commit trace $m(t)$, and fixed latch trace $\ell(t)$ over $[0, T_{\mathrm{stop}}]$,*

$$\frac{\partial E_{\mathrm{ctrl}}^{(\textsc{Confidence-Gated})}}{\partial G_{\mathrm{conf}}} = \int_0^{T_{\mathrm{stop}}} (0.25 + 0.75 m(t))\, v_{\mathrm{store}}(t)^2 (1 - \ell(t))\, dt > 0. \tag{96}$$

Proof. Equation (66) is affine in G_{conf}:

$$E_{\mathrm{ctrl}}^{(\textsc{Confidence-Gated})} = G_{\mathrm{scout}} \int_0^{T_{\mathrm{stop}}} v_{\mathrm{store}}(t)^2 (1 - \ell(t))\, dt + G_{\mathrm{conf}} \int_0^{T_{\mathrm{stop}}} (0.25 + 0.75 m(t))\, v_{\mathrm{store}}(t)^2 (1 - \ell(t))\, dt. \tag{97}$$

Differentiating with respect to G_{conf} immediately yields the stated integral. The integrand is nonnegative because $m(t) \in [0, 1]$, $v_{\mathrm{store}}(t)^2 \geq 0$, and $1 - \ell(t) \geq 0$. It is strictly positive on every successful startup interval because $v_{\mathrm{store}}(t)$ is positive over a set of nonzero measure while the latch is still open, and the factor $(0.25 + 0.75 m(t))$ is bounded below by 0.25. □

Parameter	Nominal value	Appears in	Monotone effect on the implemented controller
$R_{\mathrm{scout}}C_{\mathrm{scout}} = \tau_{\mathrm{s}}$	0.75 ms	Eq. (11)	larger value smooths the scout margin more heavily before confidence is computed
$R_{\mathrm{conf}}C_{\mathrm{conf}} = \tau_{\mathrm{c}}$	50 ms	Eq. (14)	larger value delays commitment by Proposition 2
Θ	0.08	Eq. (15)	larger value delays or suppresses commitment by Proposition 2
V_{floor}	10 mV	Eq. (13)	larger value postpones confidence accumulation until scouts have larger amplitude
G_{conf}	8 nS	Eq. (66)	larger value raises pre-handoff control energy linearly by Proposition 3
G_{TC}	18 nS	TC-RANKEDcontrol branch	smaller value lowers control burden but does not create abstention behavior

Table 7: Hyperparameter sensitivity summary for the implemented controller family. The table is analytical, not fitted: every statement follows from the explicit netlist equations proved in the main text and appendix.

B Additional Experimental Details

Table 8 gives the exact DEEPEN manifest. The broader falsifier suite is summarized after it.

Case	Family	(V_A, V_B) (mV)	Ratio	Delay of B (s)	Active designs
`static_r1_v101`	static	$(100, 101)$	$1:1$	0	CONFIDENCE-GATED, BLIND PACKET, TC-RANKED
`static_r1_v103`	static	$(100, 103)$	$1:1$	0	CONFIDENCE-GATED, BLIND PACKET, TC-RANKED, BLIND MERGE
`static_r1_v105`	static	$(100, 105)$	$1:1$	0	CONFIDENCE-GATED, BLIND PACKET, TC-RANKED
`static_r3_v101`	static	$(100, 101)$	$1:3$	0	CONFIDENCE-GATED, BLIND PACKET, TC-RANKED

Case	Family	(V_A, V_B) (mV)	Ratio	Delay of B (s)	Active designs
`static_r3_v103`	static	$(100, 103)$	$1:3$	0	Confidence-Gated, Blind Packet, TC-Ranked, Blind Merge
`static_r3_v105`	static	$(100, 105)$	$1:3$	0	Confidence-Gated, Blind Packet, TC-Ranked
`late_r1_v101`	late	$(100, 101)$	$1:1$	4	Confidence-Gated, Blind Packet, TC-Ranked
`late_r1_v103`	late	$(100, 103)$	$1:1$	4	Confidence-Gated, Blind Packet, TC-Ranked, Blind Merge
`late_r1_v105`	late	$(100, 105)$	$1:1$	4	Confidence-Gated, Blind Packet, TC-Ranked
`late_r3_v101`	late	$(100, 101)$	$1:3$	4	Confidence-Gated, Blind Packet, TC-Ranked
`late_r3_v103`	late	$(100, 103)$	$1:3$	4	Confidence-Gated, Blind Packet, TC-Ranked, Blind Merge
`late_r3_v105`	late	$(100, 105)$	$1:3$	4	Confidence-Gated, Blind Packet, TC-Ranked

Table 8: Exact DEEPEN manifest implemented by `tools/run_h1_deepen.py`. All cases use same polarity and a ramp of $10\,\mathrm{mV\,s^{-1}}$.

Case	Stress class	Polarity	Ratio	Key overrides
`fa_001`	source collapse	same	$1:5$	$R_{\text{load}} = 1\,\mathrm{M\Omega}$, source A collapses to 5% at 4.2 s
`fa_002`	source collapse	mixed	$1:20$	source A collapses to 5% at 4.4 s

Case	Stress class	Polarity	Ratio	Key overrides
`fa_003`	polarity mismatch	mixed	1 : 20	no collapse; low-voltage mixed-polarity stress
`fa_004`	UVLO chatter	same	1 : 1	$(V_{\mathrm{handoff}}, V_{\mathrm{handoff,fall}}) = (0.8\,\mathrm{V}, 0.7\,\mathrm{V})$, delayed dual collapse, $R_{\mathrm{load}} = 100\,\mathrm{k\Omega}$
`fa_005`	mixed conflict	mixed	1 : 1	$R_{\mathrm{load}} = 50\,\mathrm{k\Omega}$, source A collapses fully then recovers
`fa_006`	dual collapse	mixed	1 : 20	heavy load, both sources collapse fully at different times
`fa_007`	unequal V_{OC}	same	1 : 5	$(V_A, V_B) = (20\,\mathrm{mV}, 50\,\mathrm{mV})$
`fa_008`	unequal V_{OC}	mixed	1 : 1	$(V_A, V_B) = (100\,\mathrm{mV}, 300\,\mathrm{mV})$
`fa_009`	leak path	same	1 : 5	router leak forced to $1\,\mathrm{k\Omega}$
`fa_010`	leak path	mixed	1 : 20	router leak forced to $1\,\mathrm{k\Omega}$

Table 9: Expanded falsifier suite used to establish the context boundary before DEEPEN.

C Implementation and Reproduction Notes

The full regeneration sequence used for the final paper is:

1. `python3 tools/run_h1_matrix.py run -designs champion,fixed,nonaware,source_blind,time_consta`
2. `python3 tools/run_h1_falsifier.py run -designs champion,fixed,nonaware,source_blind,time_con`
3. `python3 tools/run_h1_robustness.py -samples 24`
4. `python3 tools/run_h1_deepen.py`
5. `python3 tools/analyze_h1_results.py`
6. `pdflatex research_paper.tex`
7. `bibtex research_paper`
8. `pdflatex research_paper.tex`
9. `pdflatex research_paper.tex`

The Blind Packetfalsifier and robustness rows in this final sequence were regenerated after repairing the deck so that `RLOAD` honors the manifest-level `R_LOAD` override. Earlier blind-heavy falsifier claims are intentionally discarded.

All claim-bearing numbers in the paper were taken from the generated CSV or JSON summaries rather than transcribed manually from raw logs. The key generated figures used in the paper are:

- `figures/h1_topology_contrast.pdf`,
- `figures/h1_primary_matrix_heatmap.pdf`,
- `figures/h1_falsifier_boundary.pdf`,
- `figures/h1_metric_accounting.pdf`,
- `figures/h1_deepen_confidence_tradeoff.pdf`,
- `figures/h1_robustness_ci.pdf`.

D Novelty-Boundary Audit Notes

The repository that launched this run began from a deliberately broad natural-language task. As a result, the earliest automated watchlist included several nontechnical false neighbors: *Leading change* (organizational change), *Fostering STEAM through challenge-based learning, robotics, and physical devices* (education review), *Artificial Intelligence, Cognitive Robotics and Nature of Consciousness*, and *The north wing of the Musin-Pushkin estate in Moscow*. These items were retained only as audit artifacts showing why the seed query had to be cleaned. They are not part of the technical prior-art boundary and were not used to justify the paper's claim.

The actual novelty boundary is therefore the one stated in Table 1 and in Section 2: strong overlap exists with adaptive dual-source startup and autonomous multi-input PMUs, but the recovered overlap set did not reveal a close same-family abstention-like pre-handoff controller. That sentence is intentionally scoped. It is a retrieval result for the recovered overlap set, not a field-wide absence proof.

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